COSMOS 954: ISSUES OF LAW AND POLICY +

Stephen Gorove *

It is unlikely that the circumstances surrounding the recent crash of Cosmos 954 over Canadian territory would have been the subject matter of a hypothetical case involving questions of liability resulting from activities in space. Situations that life produces frequently surpass the wildest imagination of man.

The facts which have been widely reported in the press[1] reveal that parts of a Russian Satellite, identified as Cosmos 954, which was believed to have been on a naval reconnaissance mission, carrying a nuclear reactor with highly radioactive materials to provide power for its remote sensing and communications systems, crashed on January 27, 1978 over a remote area in Canada. Due to an apparent malfunction, Soviet flight controllers were unable to send the satellite into a higher and at least temporarily safer orbit or to direct it to re-enter over some desolate ocean area.[2]

The Soviet program counted on the satellite burning up and disintegrating high in the atmosphere, where its radioactive cargo could presumably disperse harmlessly. However, the accident in Canada revealed that some dangerous material could reach the ground.

While there was radiation emitted from the discovered debris, there were no reported cases of injury to persons or specific damage to property.[3] Nonetheless, the risk of potential danger and public health hazards prompted certain precautionary measures by the Canadian government involving a large number of scientists, technicians and military personnel and about a dozen aircraft—many of them from the United States—in search and recovery operations relative to the Russian satellite.[4]

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*Chairman of the Graduate Program of the School of Law and Professor of Law, University of Mississippi Law Center. Author, Studies in Space Law: Its Challenges and Prospects (1977).

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[1]See, for instance, issues of the N.Y. Times, January 25, 26 and 27, 1978.

[2]*Id.*, April 2, 1978, p. 10-L.

[3]U.N. Doc A/AC.105/217; N.Y. Times, April 2, 1978, p. 10-L.

[4]*Id.*, February 5, 1978, p. 6-L; *id.*, April 2, 1978, p. 10-L.

Canadian officials estimated that the costs of the search operations had already exceeded several million dollars, not including the expense of a significant number of American scientists and engineers who had taken part.[5]

The Canadian government indicated that it would seek reimbursement for some of the costs from the Soviet Union.[6] Though the Soviet Union has acknowledged that a satellite, Cosmos 954, dropped out of the sky over Canada, the Russians have not been anxious to concede ownership of the fragments that have been discovered.[7]

The real significance of Cosmos 954 is that this was the first nuclear-related crisis involving space activities. As such, the accident started people to think more seriously about the problem of the control of nuclear energy particularly in space. At the same time, the accident brought up questions pertaining to liability. Should a nation be internationally liable for any costs which may be incurred as a result of precautionary measures taken by governments in connection with potential threats of atomic radiation? More specifically, would the costs of scouting for radioactive contamination, involving the deployment of a number of planes by Canada and the expenditures in connection with the use of scientific and other personnel be a basis of a damage claim? Should Canada be entitled to compensation or reimbursement from the Soviet Union for such costs? Should Canada be obligated to return to the Soviet Union any parts of the satellite which were recovered? Should there be new international agreements concluded governing the use of atomic energy in space?

Whatever answers may be given to the above questions, it has been observed that the mounting costs of the complicated effort, involving logistical problems made worse by the sub-Arctic cold, make the question of liability and the extent of damages much more intricate.[8]

The Soviet Union has indicated its willingness to pay for the damages resulting from the fall of the satellite, but it was said there was more at issue than just the question of damages.[9]

Obviously, the Soviet incident may also raise some hypothetical questions. What if the Russian satellite had landed close to a big, populated city and massive hysteria developed requiring hospitalization of people for shock or mental anxiety? What if

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[5] *Id.*, April 2, 1978, p. A-10.

[6] *Ibid.*

[7] N.Y. Times, January 25, 1978, p. A-11.

[8] *Id.*, January 31, 1978, p. A-11; *id.* February 8, 1978, p. A-9.

[9] *Id.*, February 15, 1978, p. A-7.

some radiation injury had occurred requiring subsequent and extensive precautionary measures? Would the cost of such measures be covered in addition to damage for radiation injuries?

In response to these questions, first of all the applicable law should be examined to determine the question of liability. Second, if it is found that the existing law is inadequate in whole or in part, the policy options ought to be reviewed in relation to what the law should be or the type of measures that ought to be taken in order to fill the gap or reduce the uncertainty in the legal framework.

I. COSMOS 954 AND THE "LEX LATA"

There are at least four international conventions with respect to which the question of liability arising out of the crash of Cosmos 954 requires analysis: the Liability Convention,[10] the Outer Space Treaty,[11] the Rescue and Return Agreement[12] and the Vienna Convention on Civil Liability for Nuclear Damage.[13]

1. Liability Convention

Under the Liability Convention the launching state is liable for damage caused by its space object.[14] The definition makes it clear that there are three elements which must be proved before recovery can be had. First, there must be damage; second, there must be a space object; and third, the damage must be caused "by" the space object. Proof of negligence is not required since the launching state is absolutely liable—irrespective of any fault on its part—for damage caused on the surface of the earth or to aircraft in flight.[15]

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[10]Convention on International Liability for Damage Caused by Space Objects, March 29, 1972 [1973] 24 U.S.T. 2389, T.I.A.S. 7762 (hereinafter cited as Liability Convention).

[11]The Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies (hereinafter referred to as the Outer Space Treaty) was signed on January 27, 1967, and entered into force October 10, 1967. [1967] 18 U.S.T. 2410, T.I.A.S. 6347, 610 U.N.T.S. 205.

[12]Agreement on the Rescue of Astronauts, the Return of Astronauts, and the Return of Objects Launched into Outer Space, April 22, 1968 [1968] 19 U.S.T. 7570, T.I.A.S. 6599, 672 U.N.T.S. 119 (hereinafter cited as Rescue and Return Agreement or Rescue Agreement).

[13]Art. I(a) of the Convention.

[14]Art. II.

[15] *Ibid.*

a. The Meaning of Damage

There are a number of questions which may appropriately be raised in connection with damage under the Liability Convention. First, what is the meaning of "damage"? How broadly or narrowly is the term interpreted? Does it only include direct or also consequential damage where the damage does not flow directly and immediately from the act, but only from the consequences of such act? Does it cover damage caused by atomic radiation? Particularly does it cover costs arising out of precautionary measures taken to prevent actual damage to people and property? Furthermore, does damage cover only physical injury or also injury affecting mental or social well-being? Specifically, does it include mental anguish, pain and suffering and other types of nonphysical damage?[16]

While the Liability Convention does not in itself give a clear answer to any of these questions, it does provide some general guidelines with respect to the meaning of damage when it equates the latter with "loss of life, personal injury or other impairment of health" on the one hand and "loss of or damage to property of states or of persons, natural or juridicial, or property of international intergovernmental organizations" on the other hand.[17]

The broad phrase, "loss of life,, personal injury or other impairment of health" may mean not only physical injury but also injury affecting mental as well as social well-being. In support of this position attention may be called to the definition used by the World Health Organization, which describes health as "a state of complete physical, mental and social well-being."[18] Viewed against this definition, the phrase "other impairment of health" seems broad enough to cover personal injury resulting in the impairment of mental faculties.[19] If health in such a sense is directly affected, recovery may be had.

Injury or damage caused by atomic radiation or fallout emanating from man-made atomic power sources—while not separately covered—would appear to be covered by the general phrase of "loss of life, personal injury or other impairment of health" and the additional phrase "loss of or damage to property." At the same time, loss of the use of property may be regarded as covered by the phrase "loss of or damage to property" if broadly interpreted.[20]

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[16]S. Gorove, Studies in Space Law: Its Challenges and Prospects 124 (1977).

[17]Art. I.

[18]Preamble to the Constitution of the World Health Organization. For a text, see WHO, Off. Rec., No. 2, pp. 100-9.

[19] *Op. cit. supra* note 16 at 125.

[20] *Id.* at 128.

b. Space Object and its Meaning

A space object may be defined as any man-made object originating from the Earth which is designed for use in outer space.[21] An atomic power source launched for use into outer space would be a space object. There can be little doubt that Cosmos 954 and its nuclear reactor, including component parts, were such space objects. Therefore, if such objects caused damage the launching state would be liable.

c. Damage Must be Caused "by" the Space Object

The phrase in the Liability Convention that the damage must be caused "by" the space object[22] means that there must be proximate causation between the damage and the activity from which the damage resulted. Additionally, the word "by" could also imply that the damage must be caused directly by the space object in the sense of physical damage or impact. It could also be interpreted to mean that consequential damage where the damage does not flow directly and immediately from the act, but only from the consequences of such act, under normal circumstances would not be covered by the Convention.[23] Strictly speaking, the Soviet satellite did not cause direct injury to people or damage to property. What makes the determination of liability particularly difficult is the fact that in view of the potential threat of atomic radiation steps were taken which appear to have been reasonably necessary under the circumstances to prevent or mitigate damages. While domestic courts may be inclined to award damages—in line with the rule of mitigation of damages and commensurate with the cost of reasonably necessary precautionary measures—it is doubtful that such costs could be recovered under the Liability Convention.

One could perhaps almost categorically state that there was proximate causation between the precautionary measures taken and the potential threat of harm from atomic radiation. If the rule of mitigation of damages were followed in such situation, which domestic courts perhaps might be more inclined to do, recovery could be had. The only problem with this conclusion is that the Canadian expenses were incurred not in connection with the mitigation of actual damages but in connection with potential damages which could have occurred by reasonable expectation but which actually did not occur. Therefore, it appears doubtful even in domestic law that recovery could be

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[21]The requirement that the space object "originate" from the Earth seems necessary if the definition is to be in line with the space treaties currently in force. This becomes apparent from Art. 5, par. 1 of the Rescue and Return Agreement which speaks of a space object's "return to Earth" and an object must have been on Earth before it could return.

Insofar as parts of a space object are concerned Art. 1(c) of the Liability Convention makes it clear that the phrase "space object" includes "component parts of a space object as well as its launch vehicle and parts thereof."

[22]Art. II.

[23]*Cf.* U.S. Sen. Comm. on Aeronautical and Space Sciences, 92nd Cong., 2d Sess., Report on Convention on International Liability for Damage Caused by Space Objects 24 (Comm. Print, 1972).

had in such a situation and it is even more doubtful that recovery could be had under the Liability Convention.

d. Justice, Equity and Restitution

The case of Cosmos 954 and the question of Liability may also be approached from the viewpoint of the Liability Convention's provision involving the principles of justice, equity and restitution. The relevant provision states that the compensation which the launching state is "liable to pay *for damage*" (emphasis added) under the Convention is to be determined "in accordance with international law and the principles of *justice* and *equity*, in order to provide such reparation in respect to the *damage* as will *restore* the person . . . to the condition which would have existed if the *damage* had not occurred."[24] (Emphasis added).

While a perfunctory reading of the foregoing provision may give the impression that the principles of justice and equity (in the sense of fairness) should determine whether or not relief would be granted and that restitution is the purpose of such relief, a closer scrutiny clearly indicates that the occurrence of actual damage is the precondition of the invocation of the principles of justice and equity as well as of the eventual restitution. This is not to say that the principles of justice and equity may or ought not be used to settle claims of a controversial nature. It just means that, at present, there is no such requirement under the Liability Convention unless there has been some actual damage in the sense of loss of or damage to property.

2. The Outer Space Treaty

There are two provisions in the Outer Space Treaty which have to be considered in connection with the question of liability. The first one stipulated that the launching state is internationally liable for damage by a space object or its component parts.[25] This provision which is contained in Article VII is couched in much more general terms than the Liability Convention and in a way constitutes *lex generalis* in contradistinction to the Liability Convention which is, to a extent, an elaboration of the principle of liability enunciated in the Outer Space Treaty much the same as the Rescue and Return Agreement is, to a great extent, an elaboration of the principles incorporated in the Outer Space Treaty concerning the rescue and return of astronauts and the return of space objects.[26]

It is doubtful that the Outer Space Treaty could be regarded as imposing more liability than the Liability Convention. This becomes quite apparent from the fact that

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[24] Art. XII.

[25] Art. VII.

[26] *Cf.* Arts. V and VIII.

the Treaty does not mention absolute liability[27] which is incorporated in the Liability Convention with respect to damage on earth or to aircraft in flight.[28] Also, the three essential elements, namely that there must be damage, that is must be done by a space object and that there must be proximate causation are equally important under the Outer Space Treaty. In addition, the Liability Convention, unlike the Outer Space Treaty, refers to principles of justice and equity in determining the compensation which the launching state is liable to pay for damage in order to provide such reparation as will restore the person on whose behalf the claim is presented to the condition that would have existed if the damage had not occurred.[29] Therefore, it is hard to see how a claimant would be better off by basing its claim on Article VII of the Outer Space Treaty rather than on the Liability Convention. Possibly, if one invoked general principles of justice and equity in the sense of fairness a case could be made out for recovery, much the same as it could under the Liability Convention if one disregarded the provisions pertaining to the requirement of actual damage. Apart from this, it may also be argued that since the Outer Space Treaty contains no definition of damage, the latter could be interpreted to include costs incurred in connection with precautionary measures against potential hazards.

Another provision which might be considered is Article VI of the Outer Space Treaty stipulating that states bear international responsibility for national activities in outer space and for assuring that such activities are carried out in conformity with the provisions of the Outer Space Treaty.

The law of international responsibility of states for national activities would certainly extend to tortious activities. However, also in such case, proof of actual damage would most likely be required.

3. The Rescue Agreement

Another convention which may be considered is the Rescue and Return Agreement. Under the provisions of this agreement there are two situations in which expenses associated with the recovery and return of space objects would have to be borne by the launching authority: first, if that authority requested the recovery of its space object and second, if the launching authority requested the return of its space object.[30] If either or both of these conditions are met, the launching authority is required to pay the expenses associated with the recovery and return.[31]

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[27] *Cf.* Art. VII.

[28] Art. II.

[29] Art. XII.

[30] Art. 5, pars. 2 and 3.

[31] Art. 5, par. 5.

4. *Vienna Convention*

Finally, it may be noted that under the Vienna Convention on Civil Liability for Nuclear Damage the operator of the nuclear facility or device is liable for damage upon proof that such damage has been caused by a nuclear incident.[32] If the fall of a space object or debris on a nuclear installation is instrumental in bringing about a nuclear incident on earth, the resulting damage may be covered under both the Liability Convention and the Vienna Convention. It should be noted, however, that proof of damage in the sense of direct damage rather than consequential damage or precautionary measures taken to avert potential damage would also appear to be required under the Vienna Convention if it is strictly interpreted.

II. POLICY OPTIONS "DE LEGE FERENDA"

A. Reduction or Elimination of Future Accidents and Hazards

The preceding survey of the questions of liability arising out of the accident involving Cosmos 954 reveals a number of uncertainties and inadequacies, if not gaps, in the legal framework. In order to reduce or eliminate similar future accidents, hazards or threats arising out of the actual or intended use of atomic power sources in space, a number of policy options may be considered individually or in combination. Among them are:

1. Conclusion of an international agreement banning the use of spacecraft carrying radioactive materials either at all or in Earth orbit, possibly in low or high orbit. If one were to outlaw simply Earth orbit situations, atomic accidents could still happen at launch or in course of the ascendency of a spacecraft. For this reason, such solution would not fully eliminate the risk of harmful radiation. Thus an all-out ban would appear preferable. However, the consequences of such ban for deep space probes will have to be weighed.

Both the United States and the Soviet Union regard nuclear devices the best power source for certain missions. The American program does not require nuclear devices for its existing orbital missions and the United States would be willing to forego the deployment of Earth orbiting satellites with such devices altogether.[33] However, they are considered essential for deep space missions where solar energy is too weak to be practical and atomic reactors have been proposed as the best power source for large satellites of the future.[34] The Soviet program, in turn, reportedly depends on atomic

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[32] Art. I(a).

[33] N.Y. Times, January 31, 1978, p. A-11; *id.*, February 14, 1978, p. A-11.

[34] *Id.*, January 27, 1978, p. A-24.

reactors in orbit to power radar systems for naval reconnaissance missions.[35] For that reason, Russia may be reluctant to accept a ban on them.

2. The second option would be to develop rigid safety precautions and standards in Earth orbiting satellites and/or the use of fail-safe devices if they can be developed and if the risks involved would be reduced to an absolute minimum. Such safety requirements may prevent any radioactive material contaminating the Earth or the lower parts of the atmosphere at launch and after launch.

3. The third alternative would not necessarily eliminate potential accidents but would serve to give advance notice of the impending danger at various stages of space operations. It would include conclusion of an international agreement on notification requirements to the effect that it would be incumbent upon the launching state to give notice of atomic power sources it intends to launch into space: prior to launching, at launching, upon descent and at impact on Earth.[36]

4. Still another option, if feasible, would be to use the shuttle to reduce or eliminate already existing or future hazards. The shuttle could retrieve active materials when their plunge into the atmosphere threatens or could assist in placing them in higher, safer orbit. The conclusion of an international agreement on such and other types of assistance should be seriously considered.

B. Options Regarding the Question of Liability

There are also several policy options which may be considered in connection with the troublesome question of liability. Among them are:

1. Conclusion of an international agreement, perhaps in the form of an additional protocal to the Liability Convention, making it clear that preventive and search measures incidental to an actual or potential threat of nuclear radiation occasioned by a space object will fall under the concept of damage as envisaged in the Liability Convention and the Outer Space Treaty. Such instrument could also include a provision to cover the costs of the indicated measures if undertaken, at the request of the threatened state, by a third party.

2. The second option would be not to conclude an additional agreement on liability incidental to preventive and search measures, but to take the position that it is covered under the Liability Convention and the Outer Space Treaty either by an expanded interpretation of the concept of damage or by a general reference to the principles of justice and equity in the sense of fairness. In this connection, it should be

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[35] *Id.*, January 31, 1978, p. A-11.

[36] A working paper containing, *inter alia*, such notification requirements was introduced at the meeting of the UN Outer Space Scientific Subcommittee earlier this year. See Doc. A/AC.105/C.2/L.115 (April 4, 1978).

observed that while a Soviet offer to pay damages may be prompted by a general desire to be fair and just so as to restore Canada to the condition that existed before the crash of Cosmos 954, any actual reimbursement of Canada by the Soviet Union for the cost of search and preventive measures may create an important precedent for any future interpretation of the concept of damage in the field of space law.

Concluding Observations

The preceding analysis of the case of Cosmos 954 has vividly brought to the fore what may be regarded as the uncertainties and inadequacies in the legal framework to cover similar situations should they arise in the future. The applicable international conventions, including the Outer Space Treaty and the Liability Convention, have not been specific about costs incurred in tracking debris and assessing public health risks. In addition, the analysis has indicated some of the policy choices that may be considered.

Admittedly, much more could be said in support of one or the other policy alternative, much the same as it could be with respect to the question of liability. For instance, even more problematical than the indicated questions of liability would be the question of payment for costs to be incurred in connection with monitoring the region of impact for a year or two to see whether any unusual amounts of radioactive material would show up in the environment. Also, the question of the cost of third party assistance—as intimated in the discussion of policy choices—remains to be considered. Whatever Canada and the Soviet Union may ultimately agree to do or not agree to do, will be an interesting final chapter in the bizarre episode of Cosmos 954.